2 Fig. 2.1 shows a simplified heat engine which consists of an insulated cylinder fitted with a perfectly fitting frictionless piston. The cylinder contains a fixed mass of an ideal gas and a heater. When heat is supplied to the gas, it can expand and does work by pushing the piston.

For Examiner's Use

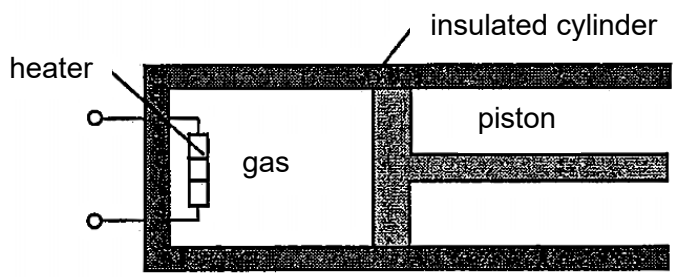

Fig. 2.1

(a) Explain in terms of the motion of the molecules in the gas, how work can be done by the heated gas to move the piston.

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..... [2]

(b) The efficiency of a heat engine is given by the following equation

$$\text{efficiency} = \frac{\text{useful mechanical work done}}{\text{input heat supplied}}$$

(i) In practice, attaining 100% efficiency is impossible. Suggest two possible reasons.

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..... [2]

(ii) Two experiments are performed on the heat engine in Fig. 2.1.

For Examiner's Use

Experiment 1
The heater provides 150 J of thermal energy with the piston held in a fixed position. The temperature rise of the gas is 30 K.

Experiment 2
The heater again supplies 150 J of thermal energy with the piston free to move so that the gas expands at a constant pressure of 1.0×10^5 Pa and does some useful work *W*. In this case, the temperature rise is 18 K and the efficiency is 40%.

1. Explain why there is a difference in the temperature rise in each case.

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…… [2]

2. In the first experiment, calculate the gain in internal energy of the gas.

gain in internal energy = …… J [1]

3. In the second experiment, calculate the work done *W* by the expanding gas.

work done = …….. J [2]

4. In the second experiment, calculate the gain in internal energy of the gas.

gain in internal energy = …… J [1]